Pavlo Melnyk

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Education

PhD in Electrical Engineering with a specialisation in Computer Vision — *Linköping, Sweden*
Computer Vision and Learning Systems, Linköping University — *Aug 2019 – Sept 2024*

- Specialisation includes Machine Learning and Geometric Deep Learning
- Funded by Wallenberg AI, Autonomous Systems and Software Program (WASP)
- Under the supervision of Michael Felsberg at the Computer Vision and Learning Systems and WASP Graduate School
- 5-year degree with 30% of coursework (90 ECTS) and 20% of teaching and other academic duties
- Thesis "Spherical NeurO(n)s for Geometric Deep Learning"

MEng in Computer Science and Technology — *Changsha, China*
Hunan University — *Sept 2016 – June 2019*

- Master's thesis "Deep Learning for Offline Handwritten Chinese Character Recognition"

BSc in Information Security Systems (Electrical Engineering) — *Pokrovsk, Ukraine*
Donetsk National Technical University — *Sept 2012 – June 2016*

Experience

Postdoctoral Researcher — *Linköping, Sweden*
Computer Vision and Learning Systems, Linköping University — *Feb 2025 -*

- Research on geometry, including applications for materials science and human pose estimation

Researcher in a Materials Science Project — *Linköping, Sweden*
Linköping University — *Feb 2024 - Feb 2025*

- WASP-WISE AI4Science collaborative project with Mårten Wadenbäck and Jonas Björk as PIs
- Developed an equivariant ML framework to be combined with DFT in a holistic approach enabling exploration of a broad range of materials and catalytic processes, resulting in the CataLiUst initiative

Co-Founder, Co-Director — *London, UK*
Demine Foundation — *Jan 2023 - Feb 2026*

- Not-for-profit organisation developing ML-assisted tools for humanitarian demining
- ML development, data collection, and international collaborations

Co-Founder, Board Member — *Linköping, Sweden*
Ukrainska Föreningen Östergötland (Regional Ukrainian Association) ukrfo.se — *Mar 2022 -*

- **Chairman** *2023 - 2025*
- Association was awarded Vitsipps Prize 2023 for extraordinary civic efforts in Linköping Municipality

Teaching Assistant, Supervisor of Master's Theses — *Linköping, Sweden*
Computer Vision and Learning Systems, Linköping University — *Aug 2019 -*

- **1500+ hours of teaching** conducted in English and Swedish
- Course leader for the Geometric Deep Learning PhD course; laboratory sessions in Multidimensional Signal Analysis, Neural Networks and Deep Learning, and Computer Vision; lessons in Signal and Image Processing; projects in Computer Vision and Conceive-Design-Implement-Operate (CDIO)
- Supervised **23 Master's theses** conducted at companies such as Maxar, Saab, Qualcomm, Ericsson, Bosch, Wikipedia, RISE, SICK, Combitech, etc.

Doctoral Student — *Linköping, Sweden*
Computer Vision and Learning Systems, Linköping University — *Aug 2019 - Sept 2024*

Student Representative — *Changsha, China*
UNESCO Youth Forum — *May 2018*

Student Researcher — *Changsha, China*
Key Laboratory of Embedded and Network Computing of Hunan Province — *Dec 2016 – June 2019*

- Developed a SOTA CNN-based method for offline handwritten Chinese character recognition (3755 classes)

Awards & Honors

- Challenge Winner, Swedish National Hackathon — *2026*
- WASP International Postdoctoral Grant (2 years), University of Pennsylvania (not taken up) — *2025*
- Top Reviewer at NeurIPS 2023 (top 10%, 1,197 of 11,725 reviewers) — *2023*
- Honorable Mention in ICML Topological Deep Learning Challenge — *2023*
- WASP Doctoral Student Grant (5 years): Collaborative project "How to Inject Geometry into Deep Learning" — *2019*
- Award by Ministry of Science and Education of Ukraine (3 years): Recipient (1/50) of a Chinese Government Scholarship to pursue a Master's in China — *2016*
- Award by the Verkhovna Rada of Ukraine (1 year): Recipient of a two-term stipend as recognition of excellent achievements in studies — *2014*

Reviewing Service

CVPR'22, 24, 26 NeurIPS'21, 23, 25 ICLR'24, 25 ICML'26 ICCV'25 ECCV'24 WACV'24 3DV'24, 26

Stack

Python · PyTorch · TensorFlow · NumPy/SciPy · scikit-learn · W&B · Git · Slurm

Languages

Ukrainian — Native · **English** — Full professional proficiency · **Swedish** — C1 (advanced), certified (2021)
Chinese — HSK5 (advanced), certified (2019) · **Spanish** — Basic

Selected Publications

QuaMo: Quaternion Motions for Vision-based 3D Human Kinematics Capture — ICLR 2026
Cuong Le, ***Pavlo Melnyk***, Urs Waldmann, Mårten Wadenbäck, Bastian Wandt

Equivariant Modelling for Catalysis on 2D MXenes — EurIPS 2025 Workshop on SIMBIOCHEM **spotlight**
Pavlo Melnyk*, Anmar Karmush*, Ania Beatriz Rodríguez-Barrera, Mårten Wadenbäck, Michael Felsberg, Johanna Rosen, Jonas Björk*

On Learning Deep O(n)-Equivariant Hyperspheres — ICML 2024
Pavlo Melnyk, Michael Felsberg, Mårten Wadenbäck, Andreas Robinson, Cuong Le

TetraSphere: A Neural Descriptor for O(3)-Invariant Point Cloud Analysis — CVPR 2024
Pavlo Melnyk, Andreas Robinson, Michael Felsberg, Mårten Wadenbäck

Steerable 3D Spherical Neurons — ICML 2022 **spotlight**
Pavlo Melnyk, Michael Felsberg, Mårten Wadenbäck

Embed Me If You Can: A Geometric Perceptron — ICCV 2021
Pavlo Melnyk, Michael Felsberg, Mårten Wadenbäck

A High-Performance CNN Method for Offline Handwritten Chinese Character Recognition and Visualization — Soft Computing 2020
Pavlo Melnyk, Zhiqiang You, Keqin Li

See all publications on Google Scholar